

Question number	Answer	Additional guidance	Mark
1 (a)(i)	C is the correct answer • A is not correct as three of the statements are correct • B is not correct as three of the statements are correct • D is not correct as three of the statements are correct		Computer (1)

Question number	Answer	Additional guidance	Mark
1(a)(ii)	D is the correct answer • A is incorrect as blood does not become less acidic • B is incorrect as muscles do not become less fatigued • C is incorrect as pH of blood does not increase		Computer (1)

Question number	Answer	Additional guidance	Mark
1(b)	A description that includes two of the following points: - accept explanation of how proton gradient set up (1) {protons / hydrogen ions} move through {membrane protein/ protein channel / ATP synthase/ ATPase} (1) {conversion of ADP + Pi to ATP / phosphorylation of ADP} (1)	can piece together reference to movement of electrons along electron carriers releasing energy to {pump protons into intermembrane space / create electrochemical gradient} accept protons move down electrochemical gradient accept ADP + Pi -> ATP	Expert (2)

Question number	Answer	Additional guidance	Mark
1(c)	D is the correct answer - A is not the correct answer as ATP is not the first molecule to contain the radioactive oxygen - B is not the correct answer as carbon dioxide is not the first molecule to contain the radioactive oxygen - C is not the correct answer as reduced NAD is not the first molecule to contain the radioactive oxygen		Computer (1)

